LIAM MCCAUSLAND

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EDUCATION

Bachelor of Science, Computer Science | *2023 – Present*. University of Manitoba

Relevant Courses: Data Structures and Algorithms, Analysis of Algorithms, Software Development 1 & 2, Programming Paradigms

In Progress: Software Engineering 1, Databases Concepts and Usage, Distributed Computing, Computer Graphics 1

PROJECTS

Banana Clicker (TypeScript, SQL, Vite, Vitest) | Software Development II | Link

- Developed a Cookie Clicker-style web game in TypeScript where players click for bananas and purchase upgrades, with user - login and save functionality backed by a SQL database.
- Applied SOLID principles and the repository design pattern to separate game logic from data access.
- Wrote automated tests with Vitest and bundled the app with Vite.
- Documented the domain model and user interaction flows, and completed a UI usability assessment.

Battleship++ (Java) | Software Development I

- Developed a turn-based Battleship variant in Java using object-oriented design, including ship movement, special abilities, and custom game effects.
- Designed a modular architecture separating game logic, persistence, and user interface components.
- Built JSON-based save/load functionality and wrote unit tests to verify game behavior.
- Produced UML diagrams, flowcharts, and technical documentation throughout development.

Rebound: Multiplayer Roblox Game (Lua, Roblox Studio) | Link

- Recreated Valve's Ricochet as a multiplayer game, implementing projectile bouncing physics, collision handling, scoring, and round-based match state management.
- Developed the custom disc physics script that anchors the core gameplay.
- Refined gameplay balance and responsiveness through playtesting.

Automata Builder (JavaScript, JointJS, HTML/CSS) | Link

- Built an interactive finite automata editor and simulator with drag-and-drop state editing, transition linking, and self-loop rendering.

- Implemented accepting and start state visualization, labeled transitions, and keyboard-based transition editing.
- Created custom formatting utilities for mathematical symbols (Greek letters, subscripts) to support formal automata notation.

Personal Portfolio Website (React, Chakra UI, HTML/CSS, JavaScript) | Link

- Designed and developed a portfolio site showcasing projects, skills, and experience, with integrated links to GitHub repositories.

TECHNICAL SKILLS

Coding Languages: Java, JavaScript, TypeScript, SQL, HTML/CSS, Lua, C, C++, C#

Frameworks and Libraries: React, Chakra UI, JointJS, Vite, Vitest

Software Programs: Git, GitHub, IntelliJ IDEA, Visual Studio Code, Vim, Roblox Studio, Microsoft Office 365

Practices: Object-Oriented Programming, SOLID Principles, Unit Testing, UML Design, JSON Data Persistence, Debugging and Refactoring

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WORK EXPERIENCE

Physiotherapy Assistant | *August 2024 – Present* Elite Sports Injury, Winnipeg, MB

- Coordinated physiotherapist schedules by booking, cancelling, and contacting clients.
- Processed client payments on TD terminals and reconciled daily payment totals at close of business.
- Administered billing and invoicing for Military, WCB, and MPI coverage programs.
- Guided clients through physiotherapist-designed rehabilitation programs and applied heat packs and TENS machines.
- Maintained clean, organized exercise areas by disinfecting equipment and organizing weights.

Stadium Operations General Labourer | *June 2021 – August 2022* Investors Group Field, Winnipeg, MB

- Constructed event setups and teardowns for major events and rentals, including stages, podiums, pipe and drape, tables, and chairs, according to floor plans.
- Prepared football and soccer fields for game days, installing nets, goal posts, benches, advertising boards, and shelters.
- Maintained stadium grounds through groundskeeping, pressure washing, field painting, and seasonal startup.
- Collaborated with crew members to set up tailgate areas, including tents, tables, chairs, and bike racks.

VOLUNTEER EXPERIENCE

Zoo Camp Supervisor | *August 2022* Assiniboine Park Zoo, Winnipeg, MB

- Collaborated with fellow supervisors and instructors to provide attentive care and supervision for children in zoo camp activities.
- Made prompt decisions and led groups of children using strong leadership and teamwork skills.
- Communicated clearly and engagingly with children to ensure an enriching and enjoyable camp experience.